A Lightweight Transformer With Multigranularity Tokens and Connected Component Loss for Land Cover Classification

Wen Lu and Minh Nguyen

***Abstract*— Onboard land cover classification provides ever-updating land cover information, supporting various intelligent satellite applications that demand timely autonomous decision-making based on current and continuous land cover data. However, due to space, weight, and power constraints, satellites possess limited computational resources, rendering them unable to execute conventional land cover classification networks. In response to this challenge, we have designed a lightweight network for land cover classification featuring two efficient transformer attention mechanisms enhanced by multigranularity tokens. Diverging from traditional transformer attention mechanisms that solely capture token-to-token correlations at a single granularity, our approach splits the tokens into four segments and uses atrous convolutions across various dilation rates to aggregate token segments from diverse receptive fields, forming token segment combinations that encompass not only point information but also information from patches of varying sizes. These multigranularity tokens are subsequently processed through the windowed squeeze axial transformer attention (WSATA) and multigranularity bilevel routing attention (MGBRA) for feature enhancement. In another aspect, empirical observations reveal that prediction errors are more prone to manifest on land covers of small extent; however, conventional methods treat all pixels uniformly. This realization motivates us to propose a novel network-agnostic loss named connected component loss (CCL), which specifically targets small-scale land covers and their boundaries. Quantitative metrics and visual interpretations from comprehensive experiments confirm that our method attains state-of-the-art accuracy on two land cover classification datasets while exhibiting significantly faster inference speed than other lightweight networks, underscoring the practical potential of our method on embedded systems.**

***Index Terms*— Land cover classification, lightweight neural network, semantic segmentation, transformer.**

I. Introduction

LAND cover classification is a multiclass semantic segmentation task where individual pixels are classified into specific natural or man-made land cover types, such as forests, urban areas, water bodies, agricultural fields, and more. Onboard land cover classification provides ever-updating land cover information, supporting various intelligent satellite applications that demand timely autonomous decision-making based on current and continuous land cover data. For instance, onboard land cover data can be integrated into early warning systems for a range of hazards, such as landslides and coastal erosion [1]. Within the field of urban administration, satellite-based land cover applications can assist the government to promptly identify and subsequently mitigate illegal construction and environmentally detrimental activities at their inception [2]. In the agricultural sector, onboard land cover analysis enables ongoing monitoring of crop health, continuous detection of pest, and real-time optimization of irrigation and fertilization practices [3]. Meanwhile, onboard processing allows satellites to process data locally, filtering and compressing it before transmission. This significantly reduces data volume sent to Earth, saving valuable bandwidth.

Over the past few decades, there has been a significant enhancement in the spatial resolution of remote sensing imagery, transitioning from 30 m to submeter levels. As a result of the increased richness in detail and structural information within remote sensing imagery, land cover classification methods have evolved and transitioned from merely distinguishing spectral or spectral–spatial information at the pixel level [4] to extracting contextual information and spatial relationships among ground objects [5]. Convolutional neural networks (CNNs) autonomously extract relevant features from data via backpropagation. This ability enables them to discern a variety of textures, shapes, and patterns critical for distinguishing diverse land cover categories, diminishing the necessity for manual feature engineering. CNNs are designed to be translation invariant, allowing them to identify features irrespective of their location within the image. This is crucial for land cover classification because the same type of land cover can appear in different parts of the image. Due to these advantages, CNNs have become one of the mainstream methods in land cover classification tasks.

The advent of vision transformers [6] has brought about a profound revolution in the field of semantic segmentation. Unlike convolution, which is intrinsically a local operator, transformers use attention as a substitute for convolution to enable long-range context modeling, which is critical for land cover classification, as land cover patterns typically extend over extensive regions and involve intricate spatial dependencies. For example, SPRLT-Net [7] integrates a novel local transformer and a spatial partition restore module, dynamically capturing spatial attention weights and restoring sequential

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subpatches to their original spatial configuration. SPFormer [8] leverages a one-layer autoencoder for dimensionality reduction and introduces innovative modules such as CSSM-MHSP and CTM, aiming to enhance feature mapping and foster meaningful information exchange among pixels. Nonetheless, this advantage is accompanied by a drawback: the process of attention involves the calculation of pairwise token relationships across all the spatial positions, resulting in significant computational complexity and substantial memory usage. Given the considerable dimensions of remote sensing tiles and the limited onboard computational resources due to space, weight, and power constraints, satellites encounter impediments in deploying conventional CNNs or transformers.

Some semantic segmentation networks use lightweight backbones, resulting in lower hardware resource consumption compared with conventional networks. For instance, BiSeNetV1 [9] and BiSeNetV2 [10] propose a dual-pathway architecture. The first pathway focuses on spatial details using wide channels and shallow layers, while the second pathway extracts categorical semantics using narrow channels and deep layers. The output features from both the paths are fused to produce the final prediction. Deep dual-resolution network (DDRNet) [11] comprises two deep branches with multiple bilateral fusions conducted in between. ABCNet [12] replaces the quadratic complexity dot-product attention mechanism with a linear attention mechanism for global contextual information extraction to maintain accuracy without significantly escalating computational consumption. DABNet [13] uses depthwise asymmetric convolution and dilated convolution to construct a bottleneck structure, reducing parameters. LiteDepthwiseNet [14] decomposes standard convolution into depthwise convolution and pointwise convolution and removes the ReLU layer and batch normalization layer in the original 3-D depthwise convolution to reduce parameters and FLOPs. The authors of STDC [15] raised concerns regarding the efficacy of using lightweight backbones derived from image classification tasks in segmentation tasks, primarily due to the absence of task-specific design. In response, they handcrafted a fundamental module comprising one 1×1 convolution followed by a series of three 3×3 convolutions with progressively reduced channels. The resulting feature outputs from these convolutions are subsequently concatenated and used as the input for the subsequent module. MKANet [16] reduces network depth by introducing the multibranch kernel-sharing atrous convolution module for multiscale feature extraction. Nevertheless, these lightweight semantic segmentation networks adopt purely CNN architectures to achieve decreased computational costs, thereby forgoing the long-range context modeling capability provided by transformers. Furthermore, although these lightweight CNNs can be deployed on embedded systems, their speed–accuracy tradeoff remains unsatisfactory.

As an exploration toward a more cost-effective semantic segmentation network for land cover classification, we propose two efficient transformer attention mechanisms enhanced by multigranularity tokens. Diverging from the traditional transformer attention mechanisms that solely capture token-to-token correlations at a single granularity, our approach splits

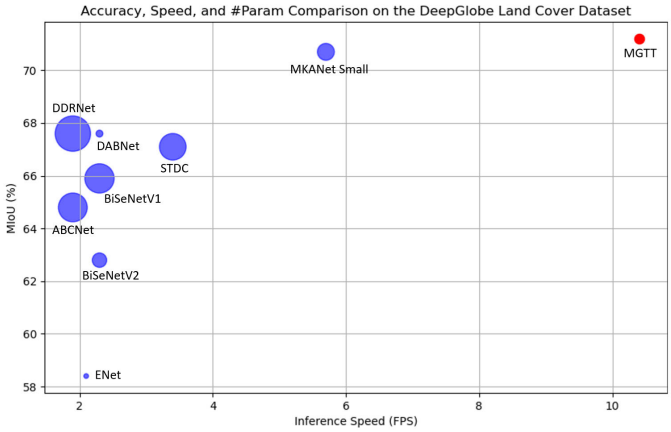


Fig. 1. Accuracy, speed, and number of network parameters' comparison on the DeepGlobe Land Cover dataset [18] with image size 2448 × 2448 pixels; the size of the circle represents the number of network parameters. Our MGTT achieves an optimal balance between speed and accuracy.

the tokens into four segments and uses atrous convolution across various dilation rates to aggregate token segments from diverse receptive fields, forming token segments combinations that encompass not only point information but also information from patches of varying sizes. These multigranularity tokens are subsequently processed through the windowed squeeze axial transformer attention (WSATA) and multigranularity bilevel routing attention (MGBRA) for feature enhancement. The WSATA is specifically designed for shallow layer features with larger dimensions. It confines the squeeze axial transformer attention to local windows, excluding remote and irrelevant information while improving computational efficiency. The MGBRA is specifically designed for deep layer features with denser semantic information. It optimizes the computational efficiency of the bilevel routing attention (BRA) [17] by eliminating the key tokens, retaining only the query and value multigranularity tokens. We also incorporate kernel-sharing atrous convolution into the encoder and integrate the two efficient transformer attention mechanisms that rely on multigranularity tokens into the decoder, forming a lightweight land cover classification network named the multigranularity tokens transformer (MGTT). As shown in Fig. 1, despite having a considerably smaller number of network parameters, MGTT attains significantly higher mean intersection over union (MIoU) accuracy than representative lightweight networks. MGTT also demonstrates significantly faster inference speed, surpassing 10 frames/s (FPS) when executed on an NVIDIA MX450 2-GB Mobile GPU (FP32: 1.67 TFLOPS), which has similar computational capacity to that of an NVIDIA Jetson Orin NX embedded system (FP32: 1.57 TFLOPS). The remarkable efficiency of MGTT highlights its practical suitability for onboard deployment.

In another aspect, empirical observations reveal that prediction errors are more prone to manifest on land covers of small extent [19]; however, conventional methods treat all the pixels uniformly. This realization motivates us to collect small-scale land covers and subsequently impose additional penalties for misclassifying on these small areas and their boundaries, thereby compelling the network to prioritize regions that pose challenges in differentiation. Its effect is evident in the second

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and fourth columns of Fig. 2, where small-scale land covers (such as ponds, rivers, small villages, minor rangeland patches, isolated barren lands, and modest farmlands) along with their immediate surroundings are picked out, while extensive forests and barren expanses are deliberately excluded.

The contributions of this work can be summarized in five aspects.

1) We design an efficient transformer attention mechanism, WSATA, for shallow layer features with larger dimensions. It confines the axial pooling and subsequent 1-D self-attention of the multigranularity tokens to partitioned windows, excluding remote and irrelevant information while improving computational efficiency.
2) We propose the MGBRA for deep layer features with denser semantic information; it optimizes computational efficiency by eliminating the key tokens, retaining only the query and value multigranularity tokens.
3) We incorporate kernel-sharing atrous convolution into the encoder and integrate the two efficient transformer attention mechanisms that rely on multigranularity tokens into the decoder, forming a lightweight land cover classification network named the MGTT.
4) We propose a novel network-agnostic loss named connected component loss (CCL), which specifically targets small-scale land covers and their boundaries.

II. Related Work

Different from convolution, which is inherently a local operator, vanilla transformer attention acquires a global receptive field by calculating pairwise token affinities across all the spatial positions. The self-attention mechanism exhibits quadratic complexity in relation to image dimensions, rendering it computationally expensive and thereby unsuitable for embedded devices, especially for high-resolution imagery inputs. Recent years have seen extensive research focused on optimizing and modifying the self-attention mechanism in vision transformer models. These efforts aim to reduce computational complexity while simultaneously maintaining or even enhancing their performance in computer vision tasks. For instance, Token slimming [20] and Mobile-Former [21] decrease complexity by reducing the number of tokens, while Twins [22], Swin Transformer [23], Shuffle Transformer [24], and HR-Former [25] limit self-attention within window partitions and lower down computational complexities to $\mathcal{O}(2C(\mathrm{HW})^{1/2}\mathrm{HW})$ and $\mathcal{O}(2\mathrm{CM}^2\mathrm{HW})$, where H and W represent the height and width of the feature map, C stands for the dimension of the tokens, and M denotes the local window size. Nevertheless, these advancements still fall short of meeting the stringent constraints imposed on satellite platforms. In this section, we present two types of attention mechanisms which are even lighter than the previously mentioned counterparts.

A. Axial Attention

Certain axial attention methods aggregate features separately along the two spatial directions. This approach enables the capture of long-range dependencies along one spatial dimension while preserving precise positional information along the other. For example, strip pooling [26] uses a band-shaped pooling window for pooling along either the horizontal or vertical dimension. Certain axial attention methods compute attention over a single axis at a time, stacking both the horizontal and vertical axial attention modules to establish a global receptive field. For instance, coordinate attention [27] factorizes channel attention into two 1-D feature encoding processes, each aggregating features along one of the two spatial directions. Subsequently, the resulting feature maps are encoded separately into a pair of direction-aware and position-sensitive attention maps. These maps are applied in a complementary manner to enhance the representations of the objects of interest within the input feature map. Axial-DeepLab [28] factorizes 2-D self-attention into two 1-D self-attentions. It achieves this by defining an axial-attention layer along the width axis of an image, which uses 1-D position-sensitive self-attention. The same definition is applied to the height-axis. Axial transformers [29] is an autoregressive model based on axial attention. This attention mechanism aligns with multiple dimensions of tensors in both the encoding and decoding settings. Criss-cross network [30] uses a novel criss-cross attention module for each pixel, enabling the harvesting of contextual information from all the pixels along its criss-cross path. Through a subsequent recurrent operation, each pixel can effectively capture dependencies across the entire image. SeaFormer [31] introduces an attention block characterized by the integration of axial squeeze and detail enhancement. This block pools input feature maps along the horizontal/vertical axis, compacting it into a column/row and performing self-attention. Subsequently, it concatenates query, keys, and values to restore detailed information lost during the squeeze, followed by feeding them into a depthwise convolution layer to enhance local details.

All these axial attentions are designed for general images with a very small spatial resolution, which results in objects being closely related in space. In contrast, remote sensing images have a spatial resolution that is several orders of magnitude higher. Pooling along the entire horizontal or vertical dimension would dilute pertinent features within middle range with irrelevant information in remote areas. Thus, it is more appropriate to capture middle-range dependencies rather than relying on the global receptive field. Our proposed axial attention mechanism is specifically designed for remote sensing imagery; it limits the axial pooling and subsequent 1-D self-attention to partitioned windows.

B. Bilevel Routing Attention

Another potential solution to mitigate the computational and memory complexities of vanilla transformer attention is to incorporate sparse attention into vision transformers, enabling each query to focus on a specific subset of key–value pairs rather than the entire set. BiFormer [17] proposes a query-aware sparsity attention mechanism, which dynamically directs each query to the most semantically pertinent subset of key–value pairs. The BRA contains a region-level routing step and a token-level attention step. The fundamental concept relies on filtering out irrelevant key–value pairs at a coarse-grained region level rather than a fine-grained token

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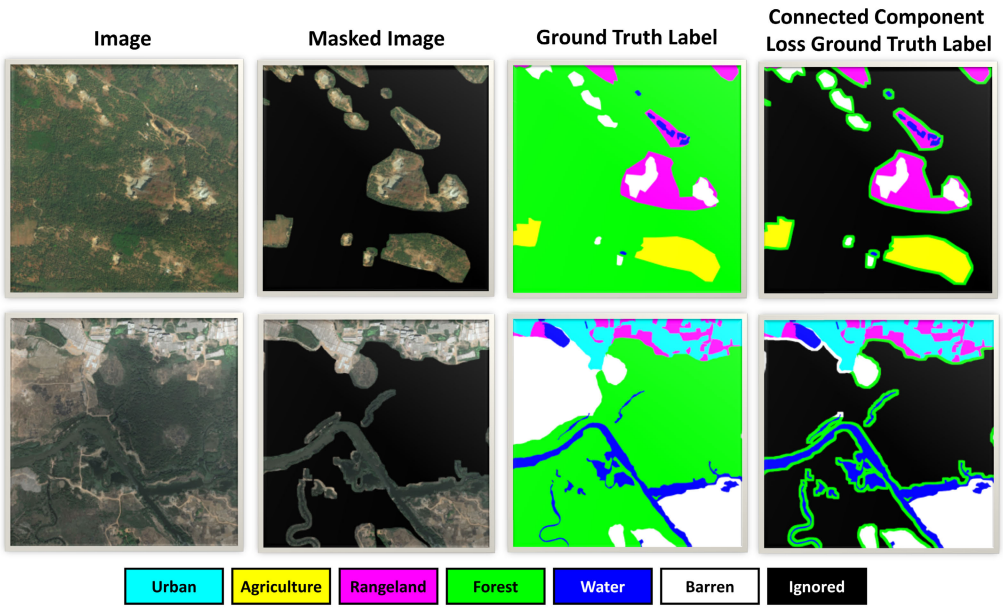


Fig. 2. Images, masked images, ground-truth labels, and CCL ground-truth labels of the DeepGlobe Land Cover dataset.

level. This is achieved through the initial construction of a region-level affinity graph, followed by pruning to retain only the top-k connections for each node. Consequently, each region solely focuses on the top-k routed regions. With the attending regions determined, token-to-token attention is applied to the key–value pairs, which are spatially dispersed.

Although the coarse region-level routing step of BRA does not incur much computation, it unavoidably results in additional GPU kernel launches and memory transactions. Consequently, due to the overheads related to kernel launch and memory bottleneck, BiFormer, which incorporates BRA as a fundamental component in the backbone, exhibits lower throughput compared with certain existing models with similar FLOPs. To adapt the BRA for embedded systems, we optimize its structure to reduce computational and memory consumption. Furthermore, we apply it selectively on deep layer features, avoiding consecutive extra GPU kernel launches and memory transactions as in BiFormer.

III. Proposed Method

This section starts with an introduction to the kernel sharing module (KSM), which serves as the encoder component for multiscale features extraction. Subsequently, it explains the WSATA and the MGBRA, which serve as decoder components to enhance the features from shallow and deep layers, respectively. Next, it describes the novel CCL, which specifically targets small-scale land covers and their boundaries. Finally, it presents the network architecture of MGTT, our proposed lightweight land cover classification network.

A. Kernel Sharing Module

Atrous spatial pyramid pooling (ASPP) [32] enables the integration of multiscale information through the use of atrous convolutions with different dilation rates. This allows the network to capture contextual information at multiple scales, facilitating better understanding and representation of complex scenes or objects. However, it also has a notable weakness. In the branch with small atrous rates, the kernels with small receptive fields are limited to learning intricate details and effectively handling small objects. Conversely, in the branch with large atrous rates, the kernels with large receptive fields can solely extract features of big objects. Insufficient communication among the branches undermines the generalizability of the individual kernels. To address this issue, the KSM, as illustrated in Fig. 3, uses multiple branches with distinct atrous rates (0, 1, 2) that collectively use a single kernel. As a result, this shared kernel can effectively scan the input features multiple times, encompassing both small and large receptive fields. Furthermore, this approach allows objects of different sizes to contribute to the training of the shared kernel, thereby augmenting the number of effective training samples and enhancing the representation ability of the shared kernel. Land objects belonging to the same class exhibit differing scales due to the varied spatial resolutions of different satellites. In addition, land covers belonging to the same class may possess distinct areas. The presence of multiscale features in remote sensing images aligns with the benefits offered by the kernel-sharing mechanism, and therefore, we incorporate the KSM in the encoder for feature extraction. The inexpensive computation cost of its kernel sharing depthwise convolutions, along with its parallel structure, significantly boosts inference speed.

B. Windowed Squeeze Axial Transformer Attention

To exclude remote and irrelevant information from axial transformer attention mechanism, we propose the WSATA, which confines the axial pooling and subsequent 1-D self-attention of multigranularity tokens to partitioned windows.

As shown in Fig. 4, WSATA first partitions the input feature map $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ into $N \times N$ nonoverlapped windows, where N is a small number, denoting the number of windows per dimension. This step is done by reshaping $\mathbf{X}$ as

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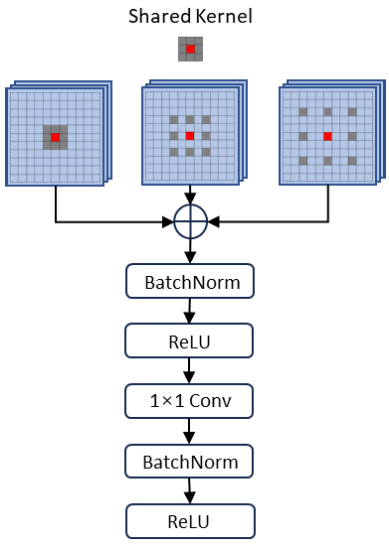


Fig. 3. Structure of the KSM.

$\mathbf{X_r} \in \mathbb{R}^{N^2\times(H/N)\times(W/N)\times C}$. Next, in each partitioned window, it pools multigranularity tokens $\mathbf{Q}$ and $\mathbf{V}$ along the horizontal or vertical axis, condensing them into a column or row, and subsequently applies self-attention within that column or row. Denote the divided feature map in an individual window as $\mathbf{x} \in \mathbb{R}^{(H/N)\times(W/N)\times C}$, the initial query tokens $\mathbf{q}'$ and value tokens $\mathbf{v}'$ are linear projections of $\mathbf{x}$

$$\mathbf{q}' = \mathbf{W_q x} \tag{1}$$

$$\mathbf{v}' = \mathbf{W_v x} \tag{2}$$

where $\mathbf{W_q}, \mathbf{W_v} \in \mathbb{R}^{C\times C}$ are learnable weights.

Following an equal division, both $\mathbf{q}'$ and $\mathbf{v}'$ are split into four token segments. No operations are conducted on the initial segment to preserve its representation of token-to-token relationships in the following multihead self-attention. Depthwise convolutions are then used on the subsequent three segments, each with distinct dilation rates (0, 1, 2). This approach aims to aggregate token segments associated with 3×3, 5×5, and 7×7 patches, respectively, to capture patch-to-patch relationships in the following multihead self-attention. After that, the four segments having varying receptive fields are concatenated to form the multigranularity query tokens $\mathbf{q}$ and multigranularity value tokens $\mathbf{v} \in \mathbb{R}^{(H/N)\times(W/N)\times C}$. Horizontal/vertical squeeze is implemented by taking average of $\mathbf{q}$ and $\mathbf{v}$ on the horizontal/vertical direction

$$\mathbf{q_h} = \frac{N}{W}\left(\mathbf{q}^{\rightarrow(C,\frac{H}{N},\frac{W}{N})}\mathbb{1}_{\frac{W}{N}}\right)^{\rightarrow(\frac{H}{N},C)} \tag{3}$$

$$\mathbf{q_v} = \frac{N}{H}\left(\mathbf{q}^{\rightarrow(C,\frac{W}{N},\frac{H}{N})}\mathbb{1}_{\frac{H}{N}}\right)^{\rightarrow(\frac{W}{N},C)} \tag{4}$$

$$\mathbf{v_h} = \frac{N}{W}\left(\mathbf{v}^{\rightarrow(C,\frac{H}{N},\frac{W}{N})}\mathbb{1}_{\frac{W}{N}}\right)^{\rightarrow(\frac{H}{N},C)} \tag{5}$$

$$\mathbf{v_v} = \frac{N}{H}\left(\mathbf{v}^{\rightarrow(C,\frac{W}{N},\frac{H}{N})}\mathbb{1}_{\frac{H}{N}}\right)^{\rightarrow(\frac{W}{N},C)} \tag{6}$$

where $\mathbf{z}^{\rightarrow(\cdot)}$ means permuting the dimension of tensor $\mathbf{z}$ as given, and $\mathbb{1}_m \in \mathbb{R}^m$ is a vector with all the elements equal to 1. Next,the 1-D multihead self-attention mechanism is applied on the horizontal/vertical squeeze tokens within each partitioned

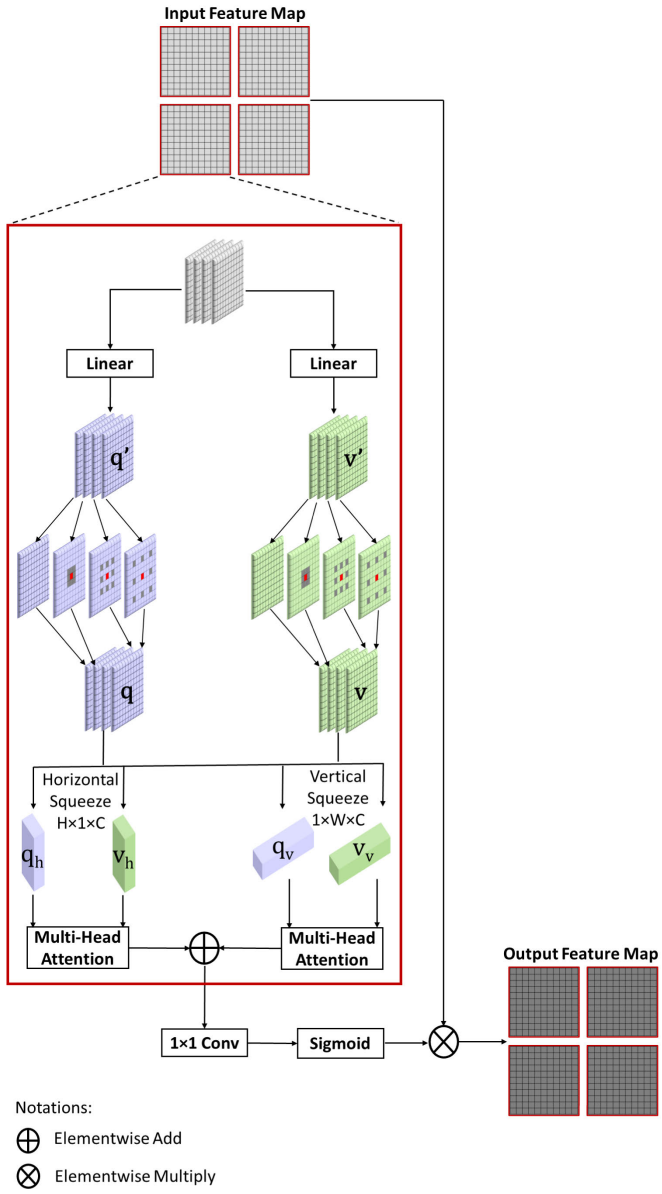


Fig. 4. Process of the WSATA.

window. We render $\mathbf{q_h}$, $\mathbf{v_h}$, $\mathbf{q_v}$, and $\mathbf{v_v}$ to be aware of their position by introducing the learnable positional embeddings $\mathbf{b_h^q}, \mathbf{b_h^v} \in \mathbb{R}^{(H/N)\times C}$, and $\mathbf{b_v^q}, \mathbf{b_v^v} \in \mathbb{R}^{(W/N)\times C}$.

The squeeze axial transformer attention mechanism in each partitioned window is expressed as (we use the single-head self-attention to replace the multihead self-attention for denotation simplicity)

$$\begin{aligned}\mathbf{y}_{i,j} = &\operatorname{softmax}\left(\left(\mathbf{q_h}_{(i)} + \mathbf{b_h^q}_{(i)}\right)\left(\mathbf{v_h} + \mathbf{b_h^v}\right)^\top\right)\mathbf{v_h} \\ &+ \operatorname{softmax}\left(\left(\mathbf{q_v}_{(j)} + \mathbf{b_v^q}_{(j)}\right)\left(\mathbf{v_v} + \mathbf{b_v^v}\right)^\top\right)\mathbf{v_v}\end{aligned} \tag{7}$$

where $(\mathbf{v_h} + \mathbf{b_h^v})^\top$ denotes the transpose of $(\mathbf{v_h} + \mathbf{b_h^v})$, and $(\mathbf{v_v} + \mathbf{b_v^v})^\top$ denotes the transpose of $(\mathbf{v_v} + \mathbf{b_v^v})$.

Subsequently, the enhanced features disperse information across channels via pointwise convolution. Following this, the sigmoid function is used to transform the features into gates, affording control over the strength of the input features via elementwise multiplication. By assigning differing degrees of

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importance to different parts of the input feature map, WSATA enables the model to selectively process and use the most pertinent information for the task.

The computational cost of linear projections and multigranularity value tokens' generation are $\mathcal{O}(2C^2\text{HW})$ and $\mathcal{O}(54\text{CHW})$, respectively. The computational complexity of squeezing the query ($\mathbf{q}$) and value ($\mathbf{v}$) is $\mathcal{O}(2\text{CN}(H+W))$, while for the axial multihead self-attention, it is $\mathcal{O}(2C(H^2+W^2))$, which is smaller than the regular multihead self-attention $[\mathcal{O}(2C(\text{HW})^2)]$ and other efficient multihead self-attentions $[\mathcal{O}(2C(\text{HW})^{1/2}\text{HW})$ or $\mathcal{O}(2\text{CM}^2\text{HW})]$. The computational consumption of the subsequent pointwise convolution and sigmoid function is $\mathcal{O}(C^2\text{HW})$ and $\mathcal{O}(\text{CHW})$, respectively, and for the final elementwise multiplication, it is $\mathcal{O}(\text{CHW})$. As a result, our proposed WSATA achieves a reduced total computational complexity of $\mathcal{O}(2C(H^2+W^2+1.5\text{CHW}))$.

Our WSATA differs from the squeeze axial attention mechanism in SeaFormer [31] in the following four aspects.

1) Diverging from the traditional transformer attention mechanisms that solely capture token-to-token correlations at a single granularity, we introduce multigranularity value tokens within the multihead self-attention, thereby modeling both token-to-token and patch-to-patch relationships.
2) To filter out remote and irrelevant information from the axial transformer attention mechanism, we incorporate a windowing scheme into the squeeze axial attention mechanism. This scheme confines the axial pooling and subsequent 1-D self-attention of multigranularity tokens to partitioned windows.
3) Unlike the multihead self-attention mechanisms in SeaFormer and other transformer models that use query, key, and value, our approach optimizes the process by eliminating the key and substituting it with value. This optimization reduces the number of parameters and computational complexity.
4) Considering the imbalanced computational complexity of the dual-path structure in SeaFormer's axial attention, we opt for the solo-path structure, abandoning the detail enhancement path. This adjustment results in faster inference speeds.

C. Multigranularity Bilevel Routing Attention

The MGBRA introduces multigranularity value tokens within the multihead self-attention and reduces computational cost by eliminating the key component and including only the query and value components in query-adaptive sparse self-attention. As illustrated in Fig. 5, the initial query tokens $\mathbf{Q}'$ and value tokens $\mathbf{V}'$ are linear projections of the input feature map $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$

$$\mathbf{Q}' = \mathbf{W_q}\mathbf{X} \tag{8}$$

$$\mathbf{V}' = \mathbf{W_v}\mathbf{X} \tag{9}$$

where $\mathbf{W_q}, \mathbf{W_v} \in \mathbb{R}^{C\times C}$ are learnable weights.

Next, it converts the initial query tokens $\mathbf{Q}'$ and value tokens $\mathbf{V}'$ into the multigranularity query tokens $\mathbf{Q}''$ and multigranularity value tokens $\mathbf{V}''$, as described in Section III-B. After that, it partitions $\mathbf{Q}''$ and $\mathbf{V}'' \in \mathbb{R}^{H\times W\times C}$ into $N\times N$ nonoverlapped windows, where N is a small number, denoting the number of windows per dimension. This step is done by reshaping $\mathbf{Q}''$ as $\mathbf{Q} \in \mathbb{R}^{N^2\times(HW/N^2)\times C}$ and reshaping $\mathbf{V}''$ as $\mathbf{V} \in \mathbb{R}^{N^2\times(HW/N^2)\times C}$. Subsequently, window-level queries and values, $\mathbf{Q_r}, \mathbf{V_r} \in \mathbb{R}^{N^2\times C}$, are derived by applying per-window average pooling on $\mathbf{Q}$ and $\mathbf{V}$, respectively. The adjacency matrix, $\mathbf{A_r} \in \mathbb{R}^{N^2\times N^2}$, of the window-to-window affinity graph is derived by matrix multiplication between $\mathbf{Q_r}$ and transposed $\mathbf{V_r}$

$$\mathbf{A_r} = \mathbf{Q_r}\mathbf{V_r}^\top. \tag{10}$$

The elements of $\mathbf{A_r}$ measure how much the two windows are semantically related. A routing index matrix, $\mathbf{I_r} \in \mathbb{N}^{N^2\times k}$, is derived by the rowwise top-k operation

$$\mathbf{I_r} = \text{topkIndex}(\mathbf{A_r}). \tag{11}$$

Consequently, the ith row of $\mathbf{I_r}$ consists of the k indices of most relevant windows for the ith window. After filtering out irrelevant values in a coarse window level, fine-grained query-to-value attention is applied in the union of the remaining windows. Specifically, for each query token in window i, it participates in self-attention calculation with all the value tokens in the union of the k routed windows indexed with $\mathbf{I_{r}}_{(i,1)}, \mathbf{I_{r}}_{(i,2)}, \ldots, \mathbf{I_{r}}_{(i,k)}$

$$\mathbf{y}_{(i,j)} = \text{softmax}\left(\mathbf{Q_{r}}_{(i,j)}\mathbf{V_{g}}_i^\top\right)\mathbf{V_{g}}_i \tag{12}$$

where $\mathbf{V_{g}}_i \in \mathbb{R}^{(\text{kHW}/N^2)\times C}$ is the gathered value tensor for the ith window.

The computational cost of linear projections and multigranularity value tokens' generation are $\mathcal{O}(2C^2\text{HW})$ and $\mathcal{O}(54\text{CHW})$, respectively. The computational complexity of the coarse window-level attention is $\mathcal{O}(2N^4C)$, while for the fine-grained query-to-value attention, it is $\mathcal{O}(2C(k/N^2)(\text{HW})^2)$. The computational consumption of the subsequent pointwise convolution and sigmoid function is $\mathcal{O}(C^2\text{HW})$ and $\mathcal{O}(\text{CHW})$, respectively, and for the final elementwise multiplication, it is $\mathcal{O}(\text{CHW})$. As a result, our proposed SBRA achieves a reduced total computational complexity of $\mathcal{O}(2C(k/N^2)(\text{HW})^2)$, which is smaller than the regular multihead self-attention $[\mathcal{O}(2C(\text{HW})^2)]$.

Our MGBRA differs from the BRA in BiFormer [17] in the following three aspects.

1) Diverging from the traditional transformer attention mechanisms that solely capture token-to-token correlations at a single granularity, we introduce multigranularity value tokens within the multihead self-attention, thereby modeling both token-to-token and patch-to-patch relationships.
2) Unlike the multihead self-attention mechanisms in BiFormer and other transformer models that use query, key, and value, our approach optimizes the process by eliminating the key and substituting it with value. This optimization reduces the number of parameters and computational complexity.
3) Instead of using a depthwise convolution, which incurs additional computational burden, for local context

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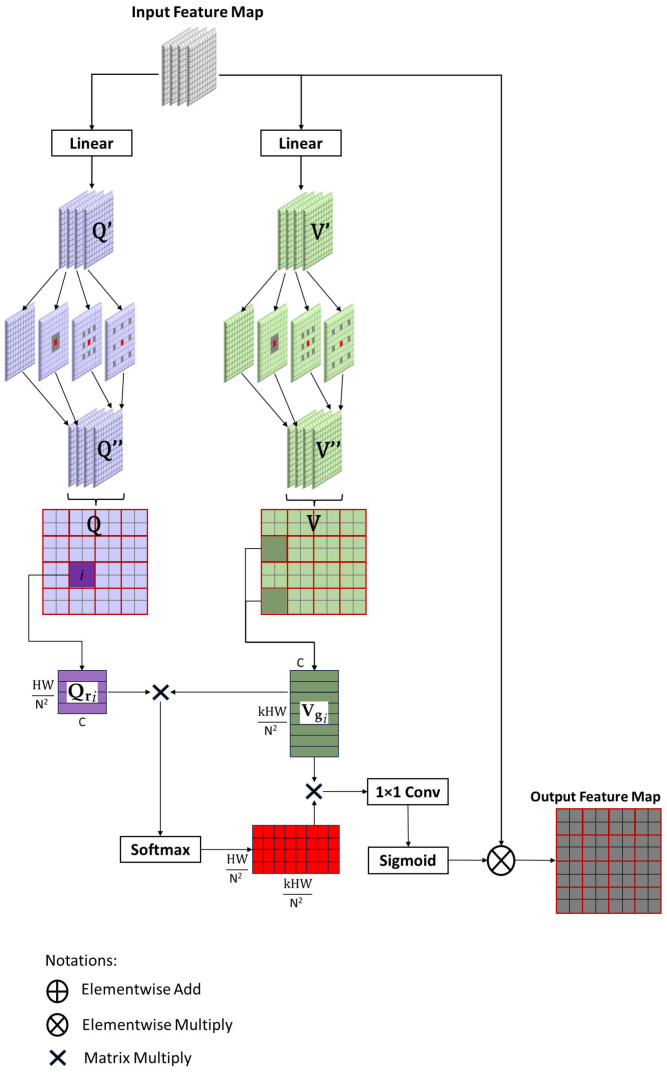


Fig. 5. Process of the MGBRA.

enhancement as done in BiFormer, we reintroduce the local context information by performing elementwise multiplication between the enhanced features and the input feature map after applying a pointwise convolution followed by the sigmoid function.

D. Connected Component Loss

Connected component labeling is an algorithm in image processing, applied for the purpose of identifying and labeling discrete connected components or objects within a binary image. The primary goal of connected component labeling is to distinguish and label distinct regions within a binary image, where pixels with the same label are part of the same object or connected component. It provides a foundation for understanding the structure of objects in a binary image and extracting meaningful information from it.

Empirical observations reveal that prediction errors are more prone to manifest on land covers of small extent. To extract the small-scale land covers and subsequently introduce additional penalties for misclassification in these smaller regions and their boundaries, we execute the CCL ground-truth generation process during the training phase, as illustrated in Fig. 6. To be specific, we initially use connected component labeling algorithm on the ground truth for land cover instance segmentation. Subsequently we filter out instances whose area proportions surpass a predetermined threshold value t. Next, we generate a binary mask for small land covers by assigning 1 to small land cover instances and 0 elsewhere. Following this, we apply a dilation operation with a predefined kernel size d on the binary mask to encompass the immediate surroundings of the small land cover instances, resulting in a dilated binary mask for small land covers. Finally, we perform elementwise multiplication of the ground truth with this mask to create the CCL ground truth.

The threshold t is a hyperparameter that determines what area proportion is defined as a small-scale land cover, and thereafter participates in the CCL calculation. The dilation kernel size d is another hyperparameter that influences the contribution of the surrounding pixels to CCL calculation. These two hyperparameters can be adjusted based on the average land cover scale present in the dataset. Fine-tuning the hyperparameters typically leads to improved results. However, to ensure generality, we refrained from fine-tuning the hyperparameters during our experiments. Instead, we used straightforward and intuitive values ($t = 0.1$, $d = 50$) to strengthen spatial detail learning and boundary recovery.

The CCL is a network-agnostic supplementary loss applicable to any semantic segmentation network. Conventional loss functions, such as cross-entropy or Dice, can be used for computing the CCL. As illustrated in Fig. 6, within the ground truth for the CCL, zero-value pixels representing large-scale land covers are relabeled as "ignored" and they are excluded from the calculation of the CCL.

E. Multigranularity Tokens Transformer

Conventional networks typically gather contextual information using multiple convolutional layers, allowing them to cover large receptive fields. Consequently, these networks possess deep architectures comprising numerous layers, resulting in slow inference speed. However, the KSM can capture a wider range of contexts for large-extent land covers and local detailed information for small-scale land covers and boundaries simultaneously. Therefore, it is not necessary to construct a deep network architecture. The architecture of our proposed lightweight land cover classification network, called MGTT, is illustrated in Fig. 7.

The output of the first KSM consists of low-level features that encompass more spatial detail information rather than semantic context. Hence, we retain these features as they are and use them for detail reconstruction and boundary recovery. The output of the second KSM comprises middle-level features with relatively small spatial resolution, sparse semantic information, and large dimension. Thus, we use the WSATA to distill the essence and subsequently enhance the features. Moreover, when compared with the simplified BRA and other transformer attention mechanisms, the WSATA exhibits a lower computational cost, making it well-suited for handling large-size features without compromising inference speed. The

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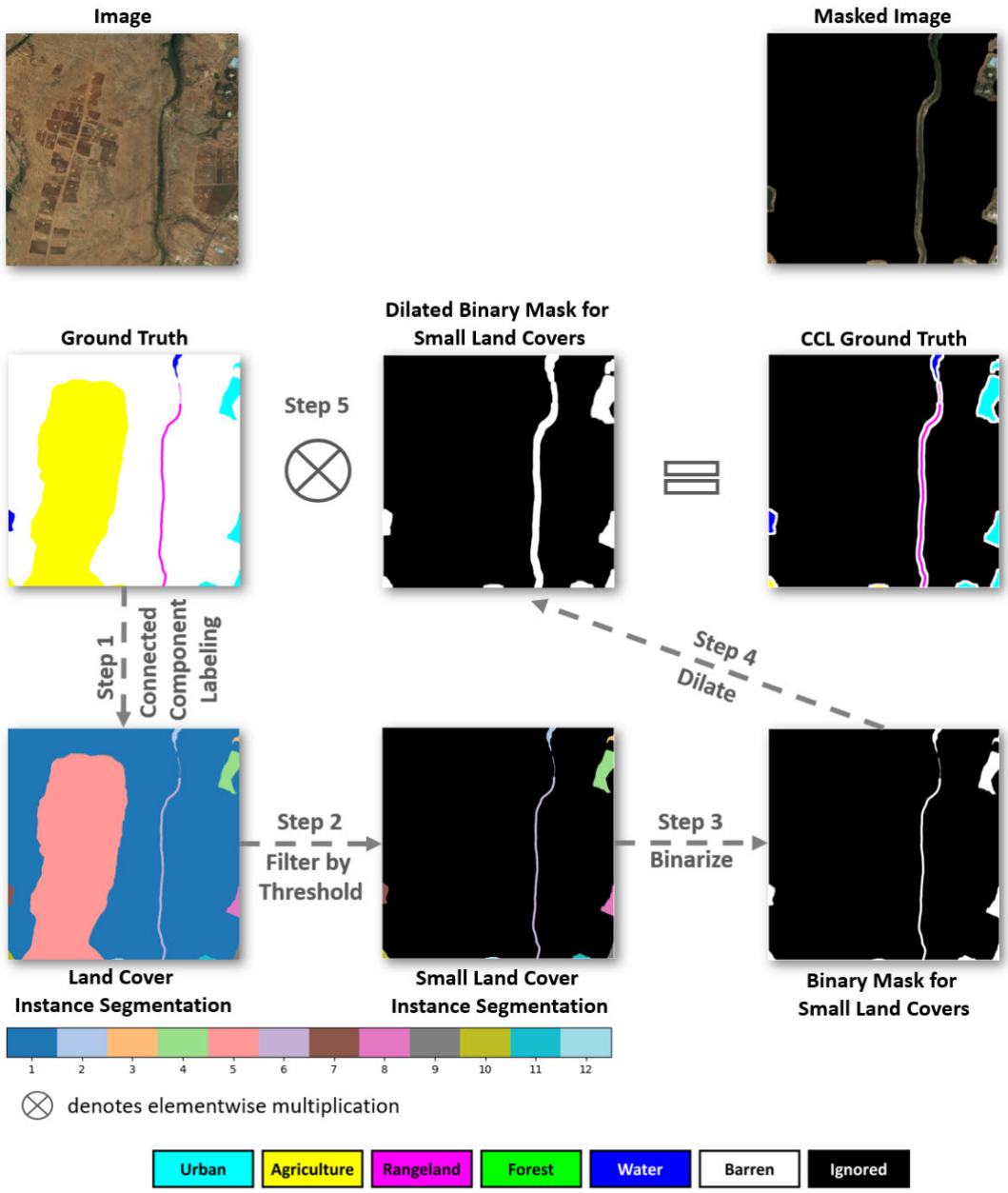


Fig. 6. CCL ground-truth generation process.

output of the third KSM contains deep-level features with relatively large spatial resolution, dense semantic information, and small dimension. Therefore, we use the simplified BRA to establish long-range dependencies and nonlocal interactions within the top-k most relevant and affinitive windows with concentrated semantic context.

Following upsampling through linear interpolation to match sizes, the features obtained from each of the three stages are concatenated. Subsequently, a pointwise convolution is used to facilitate information exchange between these features and generate class logits, serving as the semantic segmentation head. The class logits are then upsampled by a factor of 8 using linear interpolation to match the dimensions of the input image. Finally, they serve as input for both the primary semantic segmentation loss function and the supplementary CCL function.

To enhance the feature extraction capability of the encoder, three pointwise convolutions, which act as semantic segmentation heads, are incorporated atop the three KSM output features during the training phase. During the inference phase, these heads are omitted, incurring no extra computational overhead. The resulting class logits from the three auxiliary semantic segmentation heads are upsampled by factors of 8, 16, and 32, respectively, before being fed to three auxiliary semantic segmentation loss functions.

The total loss L_t is the weighted sum of the primary semantic segmentation loss L_p, the supplementary CCL L_s, and the auxiliary semantic segmentation losses L_a

$$L_t = w_1 \times L_p + w_2 \times L_s + w_3 \times L_a. \tag{13}$$

Similar to the CCL L_s, conventional loss functions, such as cross-Entropy or Dice, can be used to compute both the

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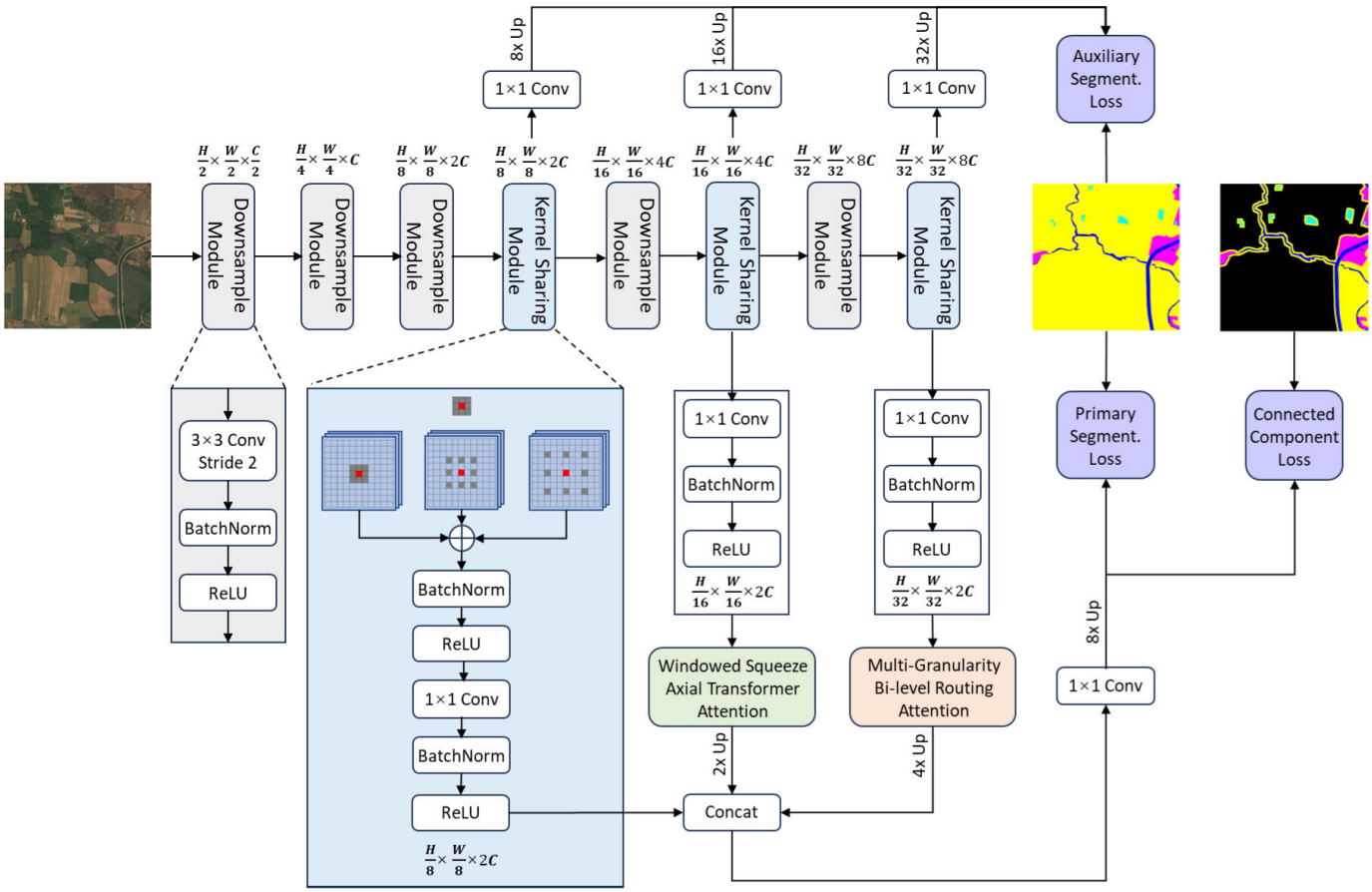


Fig. 7. Architecture of MGTT.

primary semantic segmentation loss L_p and the auxiliary semantic segmentation losses L_a. We used cross-entropy as the loss functions in the subsequent experiments, in that case, the total loss L_t is

$$L_t = -w_1 \times \sum_{i=1}^{N}\sum_{j=1}^{n} y_{ij} \cdot \log\left(p_{ij}\right) - w_2 \times \sum_{i=1}^{N'}\sum_{j=1}^{n} y_{ij} \cdot \log\left(p_{ij}\right) - w_3 \times \sum_{k=1}^{3}\sum_{i=1}^{N}\sum_{j=1}^{n} y_{ij} \cdot \log\left(p_{ij}^{k}\right) \quad (14)$$

where N is the total pixels within a batch, N' is the considered pixels (the pixels whose values equal 1 in the dilated binary mask for small land covers) for the CCL, n is the number of classes, y_{ij} is the true probability distribution over classes (often one-hot encoded), p_{ij} is the predicted probability distribution over classes by the primary head, and p_{ij}^{k} is the predicted probability distribution over classes by the auxiliary head k.

Fine-tuning the hyperparameters w_1, w_2, and w_3 typically leads to improved results. However, to ensure generality, we refrained from fine-tuning the hyperparameters in our experiments. Instead, we used straightforward and intuitive values. In the subsequent experiments, we assigned weights of $w_1 = 1$ and $w_2 = 2$ to apply a triple misclassification penalty on small-scale land covers and their boundaries, thus urging the network to give higher priority to regions that present difficulties in differentiation. Since the auxiliary losses do not directly affect prediction accuracy, we reduced their weight by assigning $w_3 = 0.2$.

IV. Experiments

A. Experimental Setting

The performance of MGTT was evaluated by conducting experiments on two satellite image land cover classification datasets: DeepGlobe Land Cover [18] and GID Fine Land Cover Classification [5].

The DeepGlobe Land Cover dataset comprises RGB satellite images of 2448 × 2448 pixels, with a pixel resolution of 50 cm. The dataset covers a total area of 1716.9 km^2 and includes six rural land cover classes. Since only the ground truths of the original competition's training set have been released, we randomly split the original training set, consisting of 803 images, into a training set, validation set, and test set at a ratio of 2:1:1, in accordance with the experimental settings in prior research.

The GID Fine Land Cover Classification dataset comprises ten Gaofen-2 satellite tiles measuring 6800 × 7200 pixels at 4-m spatial resolution, with 15 land cover classes. Due to limitations in GPU memory capacity, the ten tiles are cropped into 90 subimages measuring 2400 × 2400 pixels. Subsequently, these 90 subimages are randomly divided into a training set, validation set, and test set at a ratio of 3:1:1, following the experimental settings in previous studies.

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TABLE I
COMPARISON ON THE DEEPGLOBE LAND COVER DATASET

Method	Per-Class IoU (%)						MIoU (%)	Speed (FPS)	#Param (M)	MACs (G)
	Urban	Agricult	Range	Forest	Water	Barren				
Proportion (%)	9.4	56.8	10.2	13.8	3.7	6.1				
Heavyweight Network										
SegNet (ResNet101) [34]	67.9	80.7	41.6	73.4	75.4	62.3	66.9	OOM		
Shan et al.'s (ResNet50) [35]	72.6	82.2	41.6	72.4	75.2	62.2	67.7	OOM		
UNet (ResNet50) [36]	76.0	85.0	34.0	77.5	80.4	55.5	68.1	OOM		
PSANet (ResNet50) [37]	-	-	-	-	-	-	68.3	OOM		
DeepLabV3 (ResNet50) [32]	-	-	-	-	-	-	68.4	OOM		
EncNet (Res2Net50) [38]	-	-	-	-	-	-	68.5	OOM		
DeepLabV3+ (Res2Net50) [39]	-	-	-	-	-	-	69.1	OOM		
GCNet (ResNet50) [40]	-	-	-	-	-	-	69.1	OOM		
PSPNet (Res2Net50) [41]	-	-	-	-	-	-	69.5	OOM		
HRNet+OCR (HRNet-W48) [42]	72.4	84.1	40.9	74.0	79.9	**65.7**	69.5	OOM		
Contrastseg (ResNet101) [43]	74.2	83.0	**43.7**	75.1	78.1	64.1	69.7	OOM		
Trans-UNet (R50-ViT-Base) [44]	**77.9**	**86.8**	36.5	76.6	**83.6**	59.8	70.2	OOM		
Lightweight Network										
ENet [45]	71.1	80.9	0.4	75.4	73.1	49.4	58.4	2.1	**0.4**	42.8
ABCNet [12]	73.5	82.0	29.0	76.0	72.5	55.9	64.8	1.9	13.4	350.6
BiSeNetV1 (Resnet18) [9]	72.7	83.9	28.1	79.0	74.7	56.7	65.9	2.3	13.8	333.4
BiSeNetV2 [10]	72.5	82.2	23.5	77.9	69.4	51.0	62.8	2.3	3.4	276.3
STDC1 [15]	74.7	85.1	31.8	76.8	75.3	59.1	67.1	3.4	11.4	278.6
DDRNet23 [11]	75.0	84.5	32.6	77.8	77.3	58.5	67.6	1.9	20.2	403.3
DABNet [13]	74.4	84.7	33.5	79.0	75.6	58.6	67.6	2.3	0.8	115.0
MKANet Small [16]	76.1	86.0	39.1	**80.6**	79.3	63.0	70.7	5.7	4.6	65.1
MGTT without CCL	74.4	85.7	38.8	77.2	79.9	61.5	69.6	**10.4**	1.5	**29.9**
MGTT with CCL	75.3	86.3	40.1	79.0	82.6	64.0	**71.2**	**10.4**	1.5	**29.9**

- indicates that the per-class IoUs were not reported by the authors, and OOM denotes Out of GPU Memory.
Inference Speed and Multiply-Accumulate Operations were measured by the original image size of 2448×2448 pixels on NVIDIA MX450 2G Mobile GPU.

In this study, we define the specific network parameters for MGTT as outlined below for subsequent experiments.

1) The base channel number C is configured to 60.
2) In the multihead self-attention mechanism used in both WSATA and SBRA, the number of attention heads is set to 8.
3) For the WSATA, we specify the number of windows per dimension (N) as 2.
4) For the SBRA, the window size $[(H/N) \times (W/N)]$ is defined as 4×4, and k is assigned a value of 4.

We used cross-entropy as the loss functions and selected AdamW [33] as the optimizer, with a batch size of 12 for the DeepGlobe Land Cover dataset and six for the GID Fine Land Cover Classification dataset. We set the base learning rate at 0.001 with cosine decay and conducted network training for 900 epochs, incorporating a warmup strategy during the initial 60 epochs. To ensure a fair comparison, we trained all the networks from scratch, without prior pretraining on other datasets. As data augmentation strategy, we applied random flipping, rotation, scaling at rates (0.7, 0.8, 0.9, 1.0, 1.25, 1.5, and 1.75), cropping to 1536×1536 pixels, and color jittering operations to the input images during the training phase. During the testing phase, no data augmentation operations were used.

The performance of the networks was evaluated by the MIoU, defined as

$$\text{MIoU} = \frac{1}{N} \sum_{c=1}^{N} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{15}$$

where N represents the number of classes, and TP_c, FP_c, and FN_c denote the number of true-positive pixels, false-positive pixels, and false-negative pixels, respectively, in Class c.

Network training was conducted using an NVIDIA RTX 3090 24-GB GPU, while network inference speed was measured using an NVIDIA MX450 2-GB Mobile GPU (FP32: 1.67 TFLOPS), which has similar computational capacity to that of an NVIDIA Jetson Orin NX embedded system (FP32: 1.57 TFLOPS).

B. DeepGlobe Land Cover Dataset Experimental Results

As presented in Table I, because of insufficient GPU memory, the heavyweight networks are unable to process the original images of size 2448×2448 pixels on an NVIDIA MX450 2-GB Mobile GPU. Consequently, deploying conventional semantic segmentation networks on systems with limited hardware resources is impractical. In contrast, lightweight networks achieve a balance between accuracy and computational consumption, enabling them to handle the original image size with an acceptable decrease in MIoU. This demonstrates the superiority of lightweight networks over heavyweight networks under strict constraints on computational resources and memory capacity.

The CCL, when compared with MGTT without the supplementary loss, effectively raises the IoU for each land cover class, with minor classes benefiting the most. As a result, it leads to a 1.6% increase in MIoU without negatively impacting the inference speed. Our method not only surpasses its lightweight counterparts but also outperforms the heavyweight networks. In terms of inference speed, number of network parameters, and multiply–accumulate operations, its superiority is more obvious.

Fig. 8 demonstrates that MGTT is more effective in identifying small-scale land covers, such as rivers. This superiority can be attributed to two factors: the KSM and the CCL.

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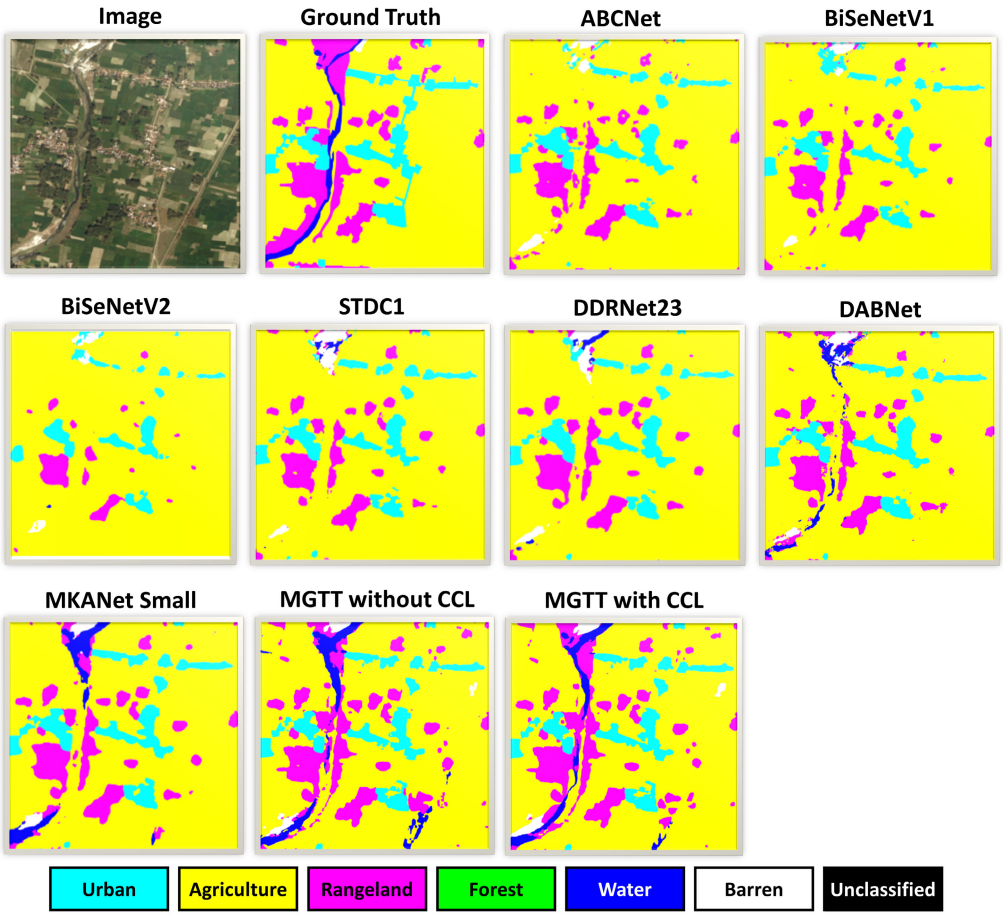


Fig. 8. Comparison of the predictions by various methods on the DeepGlobe Land Cover dataset.

The KSM has multiscale receptive fields while preserving spatial resolution, thus retaining spatial details. Even without the CCL, MGTT without CCL outperforms other networks in predicting the river, which contains more spatial details. On the other hand, the CCL can also effectively improve small-scale land cover recognition and boundary recovery, as evidenced by the prediction of MGTT with CCL, in which the river is reconstructed more accurately. This visual observation aligns with the per-class IoUs presented in Table I, where MGTT with CCL significantly outperforms other networks in recognizing small-scale land covers such as water.

C. GID Fine Land Cover Classification Dataset Experimental Results

The GID Fine Land Cover Classification dataset presents significant challenges due to its limited sample size (only ten tiles measuring 6800 × 7200 pixels) and highly imbalanced distribution of land cover classes. In this dataset, three land cover classes have proportions that are as low as less than 1%. We selected this dataset with diverse land cover classes to evaluate the discriminatory capability of our method for fine-grained and similar land cover classes. In addition, we assessed its robustness concerning highly imbalanced land cover classes.

The increasing number of land cover classes poses additional challenges to a network by testing its efficiency in terms of memory usage. According to Table II, ABCNet, BiSeNetV2, and DABNet were unable to measure the inference speed due to insufficient GPU memory. Our MGTT remains the sole network achieving an inference speed exceeding 10 FPS, placing it significantly ahead of other networks. The CCL consistently enhances the per-class IoUs of the majority of land cover classes, enabling MGTT to maintain its superiority over both the heavyweight and lightweight networks.

Due to limitations in GPU memory capacity, prior studies [50], [56], [57], [58] focusing on heavyweight CNN and transformer networks for land cover classification have conducted preprocessing on the original remote sensing images. These images were downsampled or cropped into smaller patches measuring 256 × 256 or 512 × 512 pixels, which were then used for training and testing heavyweight CNN and transformer networks. Nevertheless, the loss of spatial detail caused by compression counteracts the enhanced feature extraction capability facilitated by a heavyweight backbone. On the other hand, subdividing the images leads to a loss of long-range context information, thereby diminishing the superior modeling ability achieved through a more complex structure. Consequently, for large-sized remote sensing images, lightweight networks offer advantages and can achieve comparable, or even superior, accuracy compared with heavyweight CNN and transformer networks, as evidenced by Table II.

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TABLE II

COMPARISON ON THE GID FINE LAND COVER CLASSIFICATION DATASET

Method	Per-Class IoU (%)															MIoU	Speed
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)	(FPS)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34		
Heavyweight																	
CFPNet [46]	56.32	67.30	53.82	48.17	78.50	69.83	35.71	12.51	64.69	0.00	40.50	2.30	91.40	59.47	75.44	50.39	OOM
Vit [6]	53.40	61.36	49.76	40.43	55.66	69.60	55.19	19.28	73.03	0.00	43.84	25.02	79.54	64.90	67.45	51.21	OOM
WiCoNet(ResNet50) [47], [48]	40.81	65.01	49.47	47.28	79.99	67.21	23.29	11.87	64.49	7.11	46.79	13.81	93.29	61.17	82.15	51.46	OOM
SegFormer(MiT-B2) [49], [48]	55.52	65.74	52.20	50.55	**82.94**	74.08	36.27	26.46	69.73	6.10	43.27	7.90	92.14	78.57	77.83	55.71	OOM
Deeplabv3+ [39]	60.75	67.04	55.45	56.36	71.30	78.44	64.44	28.72	76.74	10.10	49.33	38.82	85.01	73.20	76.28	59.87	OOM
Twins [22]	61.78	68.06	57.11	59.01	70.60	80.76	71.74	23.43	78.62	0.03	54.55	45.11	85.35	75.06	77.64	61.10	OOM
Swin [23]	62.13	69.08	58.31	59.51	71.53	81.24	68.42	35.18	77.88	3.95	57.14	48.12	91.15	84.87	76.77	63.41	OOM
P-Swin [50]	63.84	69.59	59.99	61.66	73.93	81.96	**72.49**	**36.33**	78.14	11.27	57.64	45.12	93.57	**87.77**	79.61	65.13	OOM
Swin-T as backbone																	
DANet [51], [48]	58.38	67.86	48.58	38.48	78.29	73.14	39.08	16.83	66.43	3.40	36.80	8.72	91.95	72.80	78.72	53.24	OOM
SETR-PUP [52], [48]	58.94	69.00	46.15	51.09	76.64	73.83	30.64	26.98	69.79	0.00	39.13	8.35	92.47	72.87	79.92	54.33	OOM
Deeplabv3+ [39], [48]	57.36	67.85	50.41	48.51	78.96	72.89	36.11	21.79	67.08	11.21	35.55	14.07	92.55	65.83	77.18	54.36	OOM
SE-Net [53], [48]	58.01	68.19	47.32	51.16	78.83	74.41	39.76	23.14	67.64	6.02	39.46	8.93	93.50	76.02	79.84	55.24	OOM
ECA-Net [54], [48]	60.08	69.20	50.92	48.98	8.68	74.81	41.06	18.64	67.86	0.00	35.04	13.33	93.93	70.16	79.95	55.29	OOM
SETR-MLA [52], [48]	55.84	65.39	46.09	52.53	81.24	74.80	45.41	25.84	66.44	13.81	33.36	5.87	93.48	74.98	82.10	55.61	OOM
PaANet [48]	57.87	68.37	49.11	52.65	82.08	75.07	39.23	22.75	67.47	5.66	36.10	12.47	**95.90**	82.94	**83.64**	56.52	OOM
Lightweight																	
ENet [45]	49.98	56.85	45.31	1.08	0.00	67.74	29.29	0.00	86.79	0.00	62.28	0.00	38.78	47.04	3.67	32.59	1.6
ABCNet [12]	47.09	58.88	41.21	34.01	17.47	70.32	7.29	12.05	88.96	16.48	83.43	22.50	39.74	43.10	57.85	42.69	OOM
BiSeNetV1 [9]	61.09	63.63	41.75	42.56	16.70	72.72	3.73	11.27	87.82	**17.44**	84.42	45.18	64.35	50.94	57.10	48.05	2.3
BiSeNetV2 [10]	59.26	65.17	49.34	43.99	56.00	79.00	3.13	9.38	93.69	9.32	86.40	55.42	39.37	44.29	50.37	49.61	OOM
STDC1 [15]	62.77	64.07	46.37	52.91	13.22	73.72	4.97	9.35	69.36	3.73	86.27	56.94	86.15	75.78	59.21	50.99	3.3
EfficientTransformer-T [55], [48]	56.37	68.06	52.46	52.18	82.65	74.17	40.38	21.56	68.32	1.89	45.62	13.03	94.84	64.35	76.08	55.25	OOM
DDRNet23 [11]	63.57	65.11	50.43	59.66	53.26	80.12	11.60	9.87	84.83	0.00	87.22	63.23	86.82	75.92	53.11	56.32	1.9
DABNet [13]	67.52	71.08	59.64	71.10	73.86	**83.79**	24.44	11.72	**95.48**	4.15	**93.27**	56.99	80.38	63.41	66.73	61.57	OOM
MKANet Small [16]	67.42	69.68	55.82	70.91	70.12	82.67	37.41	16.44	93.82	10.98	92.30	61.39	86.34	78.90	69.70	64.26	5.6
MGTT without CCL	69.24	71.35	59.23	72.86	73.73	81.95	31.04	16.58	93.51	9.47	92.18	**65.19**	87.09	81.83	70.36	65.04	**10.1**
MGTT with CCL	**70.89**	**71.79**	**61.30**	**74.57**	74.99	83.11	30.44	18.77	95.16	5.56	93.17	64.36	87.89	85.12	72.47	**65.97**	**10.1**

OOM denotes Out of GPU Memory.
Inference speeds were measured by cropped image size of 2400×2400 pixels on NVIDIA MX450 2G Mobile GPU.

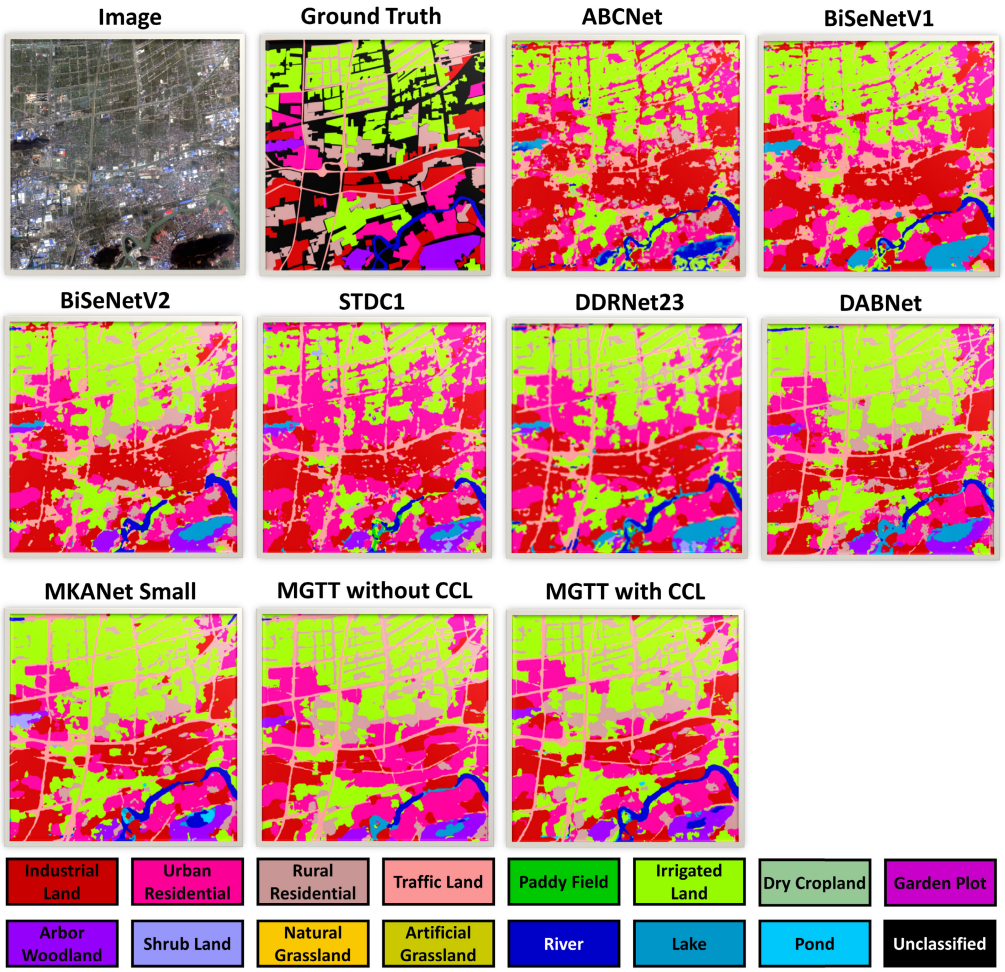


Fig. 9. Comparison of the predictions by various methods on the GID Fine Land Cover Classification dataset.

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TABLE III
INFLUENCE OF WSATA AND SBRA

Method	Per-Class IoU (%)															MIoU
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34	
MGTT lacks both	67.85	70.27	60.99	73.59	66.51	81.29	30.33	**19.83**	96.18	2.57	92.20	46.63	71.98	59.59	62.14	60.13
MGTT lacks WSATA	67.40	70.53	59.81	74.31	60.37	82.07	20.62	13.56	95.97	5.29	93.57	47.05	**90.98**	**95.74**	62.51	62.65
MGTT lacks SBRA	67.27	68.39	56.60	73.89	73.81	**83.99**	27.67	16.45	**97.70**	**7.78**	**96.10**	61.41	74.39	67.02	71.24	62.91
MGTT has both	**70.89**	**71.79**	**61.30**	**74.57**	**74.99**	83.11	**30.44**	18.77	95.16	5.56	93.17	**64.36**	87.89	85.12	**72.47**	**65.97**

TABLE IV
COMPARISON OF WSATA WITH VARIOUS NUMBERS OF WINDOWS PER DIMENSION

Method	Per-Class IoU (%)															MIoU	Speed
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)	(FPS)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34		
MGTT ($N = 0$)	64.72	68.82	**61.88**	74.05	74.84	83.11	23.07	19.98	95.11	2.50	**94.97**	**73.66**	89.06	85.10	70.80	65.44	**10.3**
MGTT ($N = 2$)	**70.89**	**71.79**	61.30	74.57	74.99	83.11	**30.44**	18.77	**95.16**	**5.56**	93.17	64.36	87.89	85.12	72.47	65.97	10.1
MGTT ($N = 4$)	67.55	69.66	61.52	**76.24**	**75.90**	**83.84**	25.76	**23.26**	94.79	0.36	94.18	69.41	**90.30**	**86.14**	**73.75**	**66.18**	10.1

Inference speeds were measured by cropped image size of 2400×2400 pixels on NVIDIA MX450 2G Mobile GPU.

TABLE V
EFFICACY OF MULTIGRANULARITY TOKENS

Method	Per-Class IoU (%)															MIoU
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34	
MGTT	**70.89**	**71.79**	**61.30**	**74.57**	**74.99**	**83.11**	30.44	**18.77**	**95.16**	**5.56**	93.17	**64.36**	87.89	85.12	72.47	**65.97**
SGTT	65.92	69.07	58.96	72.77	73.43	82.02	**31.01**	17.47	92.21	2.22	**93.36**	61.15	**92.03**	**86.51**	**75.17**	64.89

Fig. 9 demonstrates that small-scale land covers, including rivers, streets, and rural house rows, are classified with greater precision by MGTT with CCL compared with o other methods. It is noteworthy that the CCL significantly aids MGTT in rectifying the misclassification of rivers as lakes.

D. Ablation Analysis

Ablation analysis experiments were conducted on the GID Fine Land Cover Classification dataset to authenticate the efficacy of each component in our method.

1) Efficacy of WSATA and SBRA: To assess the efficacy of the WSATA and SBRA, we created three distinct network variants: one devoid of both WSATA and SBRA, another lacking only WSATA, and a third lacking only SBRA. As indicated in Table III, it is evident that both WSATA and SBRA individually contribute to an increase of over 2% in the MIoU, and their combined effect synergistically enhances the MIoU by over 5%.

2) Impact of Window Numbers in WSATA: To evaluate the impact of varying the number of windows per dimension (N) in WSATA, we constructed two variant networks: one with no window partitioning ($N = 0$) and another with four window partitionings per dimension ($N = 4$). We compared these variants with our network configuration from prior experiments, i.e., two window partitionings per dimension ($N = 2$). As indicated in Table IV, increasing the number of windows per dimension (N) consistently raises MIoU, demonstrating that by confining the axial pooling and subsequent 1-D self-attention to partitioned windows, irrelevant information can be excluded, allowing the attention mechanism to focus on pertinent factors and thus improve classification accuracy. On the other hand, Table IV also shows that augmenting the number of windows results in only a marginal rise in computational overhead. The inference speed measurements align with the computational complexity analysis in Section III-B.

3) Impact of Multigranularity Tokens: To assess the effectiveness of multigranularity tokens, we constructed an alternate network using only single-granularity tokens, denoted as SGTT. As illustrated in Table V, MGTT exhibits a superior performance, surpassing SGTT by 1.1% in terms of MIoU. This validates the premise that devising token segment combinations that incorporate not only singular point information but also features from diverse patch sizes is advantageous for multihead self-attention mechanisms.

4) Scalability of MGTT: To evaluate the scalability of MGTT, we expanded the network's breadth by augmenting the base channel number C to 80. As detailed in Table VI, the per-class IoU values for the majority of classes exhibited enhancements corresponding to the augmented channel numbers.

5) Influence of Loss Weights: To assess the influence of loss weights, an experiment was performed wherein no loss weights were used, namely, all the loss weights were uniformly set to a value of 1. As presented in Table VII, applying heavier misclassification penalty on small-scale land covers and their boundaries by increasing the weight w_2 of the supplementary CCL L_s results in heightened accuracy. Meanwhile, diminishing the weight w_3 of the auxiliary semantic segmentation losses L_a, which solely transmit gradients through the encoder, redirects the training focus toward the other two losses within the main head, which transmit gradients throughout the entire network architecture. In addition, given that the auxiliary semantic segmentation losses L_a comprise three losses originating from the three auxiliary heads, scaling down their weight w_3 becomes imperative for proportional adjustment. This experimental result validates the rationale behind our chosen loss weight configuration.

6) Influence of the Number of Training Samples: To evaluate the impact of varying training sample sizes on model performance, we partitioned the dataset randomly into distinct

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TABLE VI
SCALABILITY OF MGTT

Method	Per-Class IoU (%)															MIoU	Speed
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)	(FPS)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34		
MGTT ($C = 60$)	**70.89**	**71.79**	**61.30**	74.57	74.99	83.11	30.44	18.77	**95.16**	5.56	93.17	64.36	87.89	85.12	72.47	65.97	**10.1**
MGTT ($C = 80$)	68.14	71.35	60.91	**75.39**	**75.16**	**83.94**	**30.91**	**21.67**	94.05	**8.03**	**94.84**	**64.71**	**91.30**	**86.23**	**75.93**	**66.84**	7.2

Inference speeds were measured by cropped image size of 2400×2400 pixels on NVIDIA MX450 2G Mobile GPU.

TABLE VII
INFLUENCE OF LOSS WEIGHTS

Method	Per-Class IoU (%)															MIoU
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34	
$w_1 = 1, w_2 = 2, w_3 = 0.2$	**70.89**	**71.79**	**61.30**	**74.57**	**74.99**	**83.11**	**30.44**	18.77	95.16	**5.56**	93.17	**64.36**	87.89	**85.12**	72.47	**65.97**
$w_1 = 1, w_2 = 1, w_3 = 1$	66.92	69.51	59.49	74.25	71.62	82.75	30.34	**20.94**	**95.58**	5.53	**94.12**	64.32	**92.25**	82.45	**74.81**	65.67

TABLE VIII
INFLUENCE OF THE NUMBER OF TRAINING SAMPLES

Method	Per-Class IoU (%)															MIoU
	Indust	Urban	Rural	Traff.	Paddy	Irrig.	Crop	Gard.	Arbor	Shrub	Natur.	Artif.	River	Lake	Pond	(%)
Prop.(%)	7.26	13.96	5.59	4.74	4.38	36.39	2.86	0.82	9.04	0.3	1.68	0.83	5.65	3.16	3.34	
50:25:25	71.17	59.75	51.79	70.50	71.12	80.92	31.27	16.36	**97.84**	5.27	90.85	55.23	92.94	90.35	55.19	62.70
60:20:20	70.89	71.79	61.30	**74.57**	74.99	**83.11**	30.44	18.77	95.16	5.56	93.17	**64.36**	87.89	85.12	72.47	65.97
70:15:15	**74.62**	**84.53**	**72.06**	69.42	**80.59**	82.79	**32.77**	**25.47**	96.87	**13.26**	**96.44**	63.77	**93.88**	**94.18**	**76.97**	**70.51**

training, validation, and test sets across different ratios. Table VIII illustrates that the model's performance improves proportionally with the size of the training set.

V. DISCUSSION

Edge computing is a distributed computing paradigm that brings data processing and computation closer to the data source or "edge" of the network, rather than relying solely on centralized servers. Edge computing enables processing data closer to the satellite, minimizing the time it takes to transfer data to and from the ground station. This is critical for applications that require real-time or near-real-time data analysis and decision-making, such as early warning systems for landslides and coastal erosion, detection of illegal construction and environmentally detrimental activities, and ongoing monitoring of crop health. By performing land cover classification and subsequent intelligent applications onboard, satellite systems can reduce the amount of data that need to be transmitted to ground stations. This optimization of data transmission can lead to significant cost savings and improved overall system efficiency.

However, satellites have limited computational resources, making them challenging to run conventional semantic segmentation networks for land cover classification. In response to this challenge, this study proposes a lightweight land cover classification network featuring two efficient transformer attention mechanisms. Our proposed dual attention transformer attains an optimal balance between speed and accuracy. Assessing at the original image size of 2448 × 2448 pixels using an NVIDIA MX450 2-GB Mobile GPU, MGTT achieves an inference speed exceeding 10 FPS, significantly faster than other lightweight networks. In addition, in the experiments, MGTT obtains comparable, or even superior, accuracy compared with heavyweight CNN and transformer networks.

Due to limitations in GPU memory capacity, prior studies focusing on heavyweight CNN and transformer networks for land cover classification have downsampled or cropped original images into smaller patches measuring 256 × 256 or 512 × 512 pixels. However, cropping images may result in the loss of important contextual information. Larger scale land cover features or patterns that are visible in the full image may not be captured in smaller patches. Smaller patches might contain only partial representations of land cover types, making it challenging to accurately classify or analyze certain features that span multiple patches. The edges of cropped patches can introduce artifacts or boundary effects that affect the accuracy of land cover classification. And spatial relationships between features may be lost when working with isolated patches, which can be crucial for understanding land cover patterns. Furthermore, managing and processing a large number of small patches can be computationally intensive and time-consuming, potentially slowing down analysis tasks. And working with small patches can be less intuitive for visualization and interpretation, as it may require stitching patches together to form a coherent image. Smaller patches may not maintain precise georeferencing information, making it more challenging to relate the image data to real-world geographic coordinates. These limitations associated with using conventional semantic segmentation networks for remote sensing imagery extend MGTT's superiority and applicability from embedded systems to computers located at ground stations.

Empirical observations reveal that prediction errors are more prone to manifest on land covers of small extent; to address this issue, we rely on the connected component labeling algorithm and propose a novel network-agnostic loss that specifically targets small-scale land covers and their boundaries. Both the quantitative metric and visual interpretations have substantiated that our proposed CCL enhances spatial detail learning, facilitates small-scale land cover reconstruction, and aids boundary recovery, all without compromising inference speed.

Despite MGTT's lower hardware resource consumption relative to other networks, it falls short of achieving real-time

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processing speeds and meeting the stringent constraints imposed on satellite embedded systems. In our future research, we will investigate effective approaches, including network pruning, parallel optimization, and hardware acceleration, to adapt and deploy it for satellite embedded systems.

VI. Conclusion

As an exploration toward a more cost-effective semantic segmentation network for land cover classification, we propose two efficient transformer attention mechanisms. The first mechanism is designed for shallow layer features with larger dimensions. It confines the squeeze axial transformer attention to local windows, excluding remote and irrelevant information while improving computational efficiency. The second mechanism is designed for deep layer features with denser semantic information. It reduces the computational cost by simplifying the BRA to include only the query and value components. We also incorporate kernel-sharing atrous convolution into the encoder and integrate the two efficient transformer attention mechanisms into the decoder, forming a lightweight land cover classification network named the MGTT. MGTT attains significantly higher MIoU accuracy than representative lightweight networks, while also demonstrating an inference speed of over 10 FPS on an NVIDIA MX450 2-GB Mobile GPU (FP32: 1.67 TFLOPS), which has similar computational capacity to that of an NVIDIA Jetson Orin NX embedded system (FP32: 1.57 TFLOPS). The remarkable efficiency of MGTT highlights its practical potential for embedded systems. In this work, we also propose a novel network-agnostic loss named CCL, which effectively improves land cover classification accuracy without compromising inference speed.

Author Contributions

Conceptualization: Wen Lu and Minh Nguyen. *Data Curation:* Wen Lu. *Formal Analysis:* Wen Lu. *Funding Acquisition:* Minh Nguyen. *Investigation:* Wen Lu. *Methodology:* Wen Lu. *Project Administration:* Minh Nguyen. *Resources:* Minh Nguyen. *Software:* Wen Lu. *Supervision:* Minh Nguyen. *Validation:* Minh Nguyen. *Visualization:* Wen Lu. *Writing—Original Draft:* Wen Lu. *Writing—Review and Editing:* Minh Nguyen. All authors have read and agreed to the published version of the manuscript.

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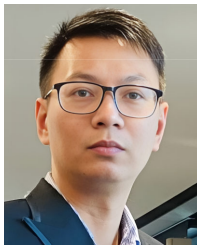
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